

# Graph Poisoning for Node Rank Manipulation

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## Abstract

Graph-based retrieval systems rely heavily on structural dependencies, making them vulnerable to adversarial manipulation. We present a black-box graph poisoning attack that degrades a target node's ranking using only edge deletions, without access to model parameters, gradients, or retraining. Prior heuristic methods treat edge influence as a static property and fail to capture how an edge's impact varies with local neighborhood structure. We address this limitation by modeling edge influence as context-dependent. Our method samples multiple ego-networks around a target node, measures empirical utility changes from edge ablations, and trains a local scorer to predict context-sensitive edge effects. At inference, predictions are aggregated across sampled subgraphs to yield stable deletion decisions. Experiments on standard benchmarks show that this approach consistently outperforms existing black-box and white-box baselines across perturbation budgets while remaining model-agnostic.

## CCS Concepts

• Information systems → Adversarial retrieval.

## Keywords

Adversarial Information Retrieval, Graph Neural Network, Graph Poisoning

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## 1 Introduction

Information retrieval systems increasingly rely on graph structures to encode relationships among queries, documents, users, and entities [9, 12]. Examples include bipartite query–document graphs that capture behavioral signals [3], nearest-neighbor graphs used for candidate generation and reranking in embedding-based retrieval,

and citation or knowledge graphs that propagate semantic relations [2]. In these settings, relevance is determined through path-based propagation or message passing, making ranking outcomes sensitive to even small structural modifications in the underlying graph [5, 10].

Given the sensitivity of these approaches to graph structure, *graph poisoning* has received growing attention by the community [6, 10]. The objective is to decrease a chosen target score in ranking or to induce a misclassification for a specific node while applying only a small number of edge deletions in the graph. Existing work can be broadly classified into four directions. *White-box* approaches [4] rely on gradients or internal losses to evaluate perturbations, which ties the attack to architectures and training signals that may not necessarily be accessible to the attacker. *Gray-box* methods [14] assume partial access to the model, such as predictions or linear approximations, but still depend on some internal knowledge to guide perturbations. *Black-box* methods [13] estimate influence through repeated interactions or reinforcement learning and can be effective, yet require large query budgets and often retraining. *Heuristic structural* methods score candidates with single snapshot statistics such as degree, centrality, or spectral measures. These methods are efficient, yet they assume that edge impacts are uniform on all graph nodes, whereas the impact of the same edge can vary depending on the target node and its neighborhood, making static global scores a weaker proxy for graph poisoning attacks.

Our work in this paper adopts a *black-box* setting in which the attacker does not have access to the model architecture, parameters, or gradients, and does not require model retraining. The attack acts only on the input graph by perturbing its edges in order to introduce downstream effects. In this setting, the effect of removing an edge depends on the neighborhood that the scorer actually uses at evaluation time, and that neighborhood can vary as indexing and approximate nearest-neighbor search change which nodes appear nearby. We treat edge influence as *conditional* on local structure. For each target, we sample budgeted  $k$ -hop ego networks that represent plausible local views, ablate candidate edges inside each view, and record the resulting change in the downstream objective of node rank demotion. Using these empirical labels, we train a scoring function on features extracted within the subgraphs to predict per-edge sensitivity. At inference, we evaluate candidate edges across a small set of sampled ego networks, aggregate the predicted effects to obtain a stable estimate, and delete the highest-scoring edges



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within the perturbation budget. This design respects the *black-box* constraints while aligning the attack with neighborhood variability.

In theory, our proposed conditional subgraph based estimation yields two benefits relative to prior poisoning strategies, namely (1) by treating edge influence as an *expectation* over plausible local neighborhoods rather than a fixed quantity tied to a single snapshot, it matches path based propagation of information used in graph representation methods and provides lower variance and greater *stability* compared to heuristic methods that assume invariant edge importance; and (2) by supervising on empirical ablations within sampled ego networks, it captures the same directional signal that gradient driven white box methods target, namely how small structural edits change scores, yet it does so without access to model parameters or losses, which keeps the method *model agnostic*. The main contributions of our work are as follows:

- A *black-box* edge-removal poisoning framework for demoting target nodes in retrieval settings, requiring no model access or retraining;
- A subgraph-conditioned training scheme that learns edge sensitivity from ego-network samples using ablation-based labels;
- An inference method that aggregates neighborhood-level predictions for robust demotion estimation;
- Extensive experiments showing consistent gains over heuristic and learning-based baselines across perturbation budgets.

## 2 Methodology

In this section, we investigate *graph structure poisoning attacks*, where an adversary is capable of *removing edges* in the original graph under a limited modification budget.

### 2.1 Problem Definition

Let  $G = (V, E)$  be a graph with adjacency  $A$  and node features  $X$ . For retrieval, query  $q$  is scored against a target node  $v_t$  by a similarity  $s(q, v_t)$ ; we write per-target utility as:

$$J(G; v_t) = \mathbb{E}_{q \sim Q(v_t)} [s(q, v_t)],$$

where  $Q(v_t)$  denotes the query distribution for  $v_t$ . The utility is the (positive) margin  $J(G; v_t) = m(v_t)$ . An attack chooses a set of incident edges  $\mathcal{B} \subseteq \mathcal{N}(v_t)$ ,  $\mathcal{N}(v_t) \triangleq \{(v_t, u) \in E\}$  (the set of edges incident to  $v_t$ ) with  $|\mathcal{B}| \leq B$  and deletes them, producing  $G' = (V, E \setminus \mathcal{B})$ . The setting is *black-box* and hence the attacker observes utilities  $J$  but has no access to model internals.

### 2.2 Proposed Method

To relate structural edits to representation changes, we analyze a linearized  $k$ -layer propagation model  $f_{\text{GNN}}(A, X) = A^k X W$ . Let “ $\cdot$ ” denote the Frobenius inner product,  $U : V = \text{tr}(U^T V)$ . Deleting a single edge induces a perturbation  $\Delta A$  and a first-order change in the embedding of  $v_t$ ,

$$\Delta z_{v_t} \approx \frac{\partial z_{v_t}}{\partial A} : \Delta A = \sum_{l=1}^k A^{l-1} \Delta A A^{k-l} X W,$$

which shows how a local edit propagates along multiple hops through powers of  $A$ . This motivates a scalar attack sensitivity

that directly measures task impact. For an edge  $e = (v_i, v_j)$  and query  $q$ , we define

$$\mathcal{S}(v_t, e | q) = \left| \frac{\partial s(q, v_t)}{\partial A_{ij}} \right|, \quad \mathcal{S}(v_t, e) = \mathbb{E}_{q \sim Q(v_t)} [\mathcal{S}(v_t, e | q)],$$

Large  $\mathcal{S}(v_t, e)$  indicates that removing  $e$  strongly decreases the utility. Closed-form computation of  $\mathcal{S}$  is intractable for deep nonlinear models, so we learn a surrogate  $f_\theta(\mathbf{h}_e) \approx \mathcal{S}(v_t, e)$  by supervised regression on context-dependent edge descriptors extracted from local subgraphs. Supervision is obtained by empirical ablation on the full graph:

$$\hat{\mathcal{S}}(v_t, e) = [J(G; v_t) - J(G \setminus \{e\}; v_t)]_+, \quad [x]_+ = \max(x, 0),$$

We adopt a larger-is-better convention for  $J$  (similarity or positive margin), so harmful deletions yield positive labels while helpful or neutral deletions map to zero. To expose structural variation while controlling cost, we generate  $m$  localized views around  $v_t$  by a stochastic  $k$ -hop expansion that samples up to  $s_h$  neighbors at hop  $h$  without replacement and then prunes to a node budget  $S_{\max}$ , i.e., retain  $v_t$  and its one-hop neighbors; if over budget, sample additional nodes from the remainder with probability proportional to  $\text{deg}^\alpha$  for  $\alpha \in [0, 1]$ , insert shortest-path connectors as needed to keep nodes reachable from  $v_t$ , and, if still over budget, drop farthest-by-hop nodes (ties by lower degree). We then denote the resulting views by  $\{G_s^{(r)}\}_{r=1}^m$ . For a given view  $G_s$ , we form an edge representation  $\mathbf{h}_e(G_s) \in \mathbb{R}^d$  only when  $e \in E(G_s)$ . The scorer  $f_\theta : \mathbb{R}^d \rightarrow \mathbb{R}_{\geq 0}$  is trained to predict the *same* label  $\hat{\mathcal{S}}(v_t, e)$  across *all* realizations in which the edge appears, using the mean-squared error objective

$$\min_{\theta} \mathcal{L}(\theta) = \frac{1}{|\mathcal{D}|} \sum_{(v_t, e, G_s) \in \mathcal{D}} \left( f_\theta(\mathbf{h}_e(G_s)) - \hat{\mathcal{S}}(v_t, e) \right)^2,$$

$$\mathcal{D} = \bigcup_{v_t} \left\{ (v_t, e, G_s) : e \in \mathcal{N}(v_t) \cap E(G_s), G_s \in \{G_s^{(r)}\} \right\}.$$

### 2.3 Inference

At inference time, we mirror training to stabilize decisions under neighborhood variability. Around  $v_t$ , we sample  $\{G_s^{(r)}\}_{r=1}^m$ , score each candidate edge in each view, and aggregate the predictions to estimate the expected effect under the subgraph distribution,

$$\tilde{\mathcal{S}}(v_t, e) = \frac{1}{m_e} \sum_{r: e \in E(G_s^{(r)})} f_\theta(\mathbf{h}_e(G_s^{(r)})), \quad m_e = |\{r : e \in E(G_s^{(r)})\}|.$$

$$\tilde{\mathcal{S}}(v_t, e) \approx \mathbb{E}_{G_s \sim \mathcal{P}(\cdot | v_t)} [f_\theta(\mathbf{h}_e(G_s))].$$

Given a budget  $B$ , we select

$$\mathcal{B}^\star(v_t) = \arg \max_{\mathcal{B} \subseteq \mathcal{N}(v_t), |\mathcal{B}| \leq B} \sum_{e \in \mathcal{B}} \tilde{\mathcal{S}}(v_t, e),$$

The aggregated score  $\tilde{\mathcal{S}}$  provides low-variance estimates of edge impact across neighborhood realizations.

## 2.4 Time Complexity Analysis

Let  $d_t$  be the degree of target node  $v_t$ ,  $R$  the number of sampled ego-networks,  $S$  the maximum subgraph size,  $C_{\text{oracle}}$  the cost of one utility evaluation  $J(\cdot)$ , and  $C_f$  the cost of a scorer forward pass.

**Training.** For each target, we compute  $J(G; v_t)$  once and  $J(G \setminus \{e\}; v_t)$  for every incident edge  $e$ , giving a labeling cost of  $O(d_t C_{\text{oracle}})$ . Sampling  $R$  subgraphs of size at most  $S$  costs  $O(RS)$ . Each edge appears in up to  $R$  subgraphs, yielding  $O(Rd_t)$  training examples and a scorer training cost of  $O(Rd_t C_f)$ . Overall training time per target is:  $O(d_t C_{\text{oracle}} + RS + Rd_t C_f)$ .

**Inference.** Attack inference mirrors training: sampling  $R$  ego-networks costs  $O(RS)$ , and scoring all incident edges in all views costs  $O(Rd_t C_f)$ . Selecting the top- $B$  edges requires  $O(d_t)$ . Total inference time per target is  $O(RS + Rd_t C_f + d_t)$ .

Our approach scales linearly with the target node’s degree and the number of sampled ego-networks, independent of global graph size beyond local structure. In contrast, state-of-the-art baselines [13] fail to scale and time out in our experiments.

## 3 Experiments

**Research Questions.** We evaluate the effectiveness, scalability, and robustness of our black-box graph poisoning method through the following research questions:

**RQ1: How does the proposed method perform on retrieval demotion compared to baselines?** We assess whether our attack degrades retrieval utility more effectively than heuristic and learning-based baselines in a fully black-box setting.

**RQ2: How scalable is the method under increasing graph size?** We analyze computational efficiency relative to other black-box methods, particularly NAG-R, by examining time and space complexity on large graphs.

**RQ3: How does adversarial effectiveness vary with perturbation budgets?** We evaluate the robustness of edge selection as deletion budgets increase, measuring whether demotion impact is sustained or improved relative to baseline trends.

### 3.1 Evaluation Setup

**Datasets.** We evaluate our attack on three standard network benchmarks: CoraFull [2], Amazon-Photo [12], and PubMed [11]. These datasets vary in size, feature dimensionality, and number of classes, providing a diverse evaluation setting. Detailed statistics for each dataset are summarized in Table 1.

**Implementation Details.** All experiments were run on an NVIDIA RTX 6000 ADA GPU with identical budgets and evaluation protocols across baselines. Methods exceeding 4 days of attack generation were labeled *OOT*. Code is available at: <https://github.com/m-hoseyny/graph-poisoning-proposed-model>.

**Baselines.** We evaluate the proposed attack against a diverse set of heuristic, unsupervised, and supervised baselines spanning white-box, gray-box, and black-box settings:

- *Degree*: A structural heuristic that removes edges incident to low-degree neighbors, assuming higher relative influence on target prominence.
- *PageRank* [7]: Ranks neighbors by personalized PageRank and removes edges to the most influential nodes, capturing long-range influence and centrality.

**Table 1: Characteristics of datasets used in our experiments.**

| Dataset           | # Nodes | # Edges | # Features |
|-------------------|---------|---------|------------|
| CoraFull [2]      | 19,793  | 126,842 | 8,710      |
| Amazon-Photo [12] | 7,650   | 238,162 | 745        |
| PubMed [11]       | 19,717  | 88,648  | 500        |

- *Random*: Deletes edges uniformly at random, serving as a control baseline.
- *VIKING* [4]: A gradient-based poisoning attack that learns edge deletions using model internals, assuming white-box access to gradients and architecture.
- *Link Prediction (VGAE)* [1]: Uses a variational graph autoencoder to estimate link probabilities and removes edges with the lowest likelihood.
- *NAG-R* [13]: A black-box attack combining Nesterov [8] Accelerated Gradient with graph rewiring to avoid local optima while preserving node degrees.
- *Nettack* [14]: A gray-box attack for targeted node misclassification that optimizes edge and feature perturbations using surrogate gradients under structural stealth constraints.

**Metric.** Attack effectiveness is measured using *Average Rank Demotion (ARD)*, which quantifies target rank degradation after attack.

### 3.2 Findings

Our experiments yield several key findings that highlight the effectiveness, efficiency, and robustness of our proposed attack strategy:

(1) As shown in Table 2, our method consistently achieves the strongest performance across all three datasets (PubMed, Amazon-Photo, and CoraFull) and both retrieval models (EPAGCL and GCA). Notably, our black-box approach outperforms gray-box attacks such as Nettack [14], despite Nettack having access to surrogate gradients and internal model information. In several cases, Nettack performs worse than simple structural heuristics like Degree or PageRank, suggesting that linear surrogate modeling fails to capture the complex non-linear interactions of modern GNNs. Similarly, VIKING [4] underperforms across all settings despite full model access, indicating that its global optimization strategy lacks the localized sensitivity needed to identify high-impact edge deletions. In contrast, our approach leverages subgraph sampling and localized edge scoring to better capture context-dependent effects. In some settings, particularly on PubMed with GCA, we observe negative rank demotions, indicating that certain edge removals unexpectedly improve target rankings. This suggests the presence of noisy or adversarial message-passing paths, whose removal can yield more coherent embeddings—an effect that is more pronounced in sparse graphs where individual edges play a larger structural role. Overall, these results address **RQ1** by demonstrating superior rank demotion performance over all baselines, including methods with internal model access.

(2) Although NAG-R is a black-box approach, it does not scale to large graphs such as CoraFull due to its quadratic time  $O(T \times B \times N^2)$  and memory  $O(N^2)$  complexity from dense adjacency operations. Its main bottlenecks include repeated cloning of dense adjacency

**Table 2: Attack performance under budget constraints ( $B=1-5$ ) on EPAGCL and GCA across three datasets (best in bold).**

| Dataset  | Method          | EPAGCL       |              |              |              |              | GCA          |              |              |              |              |
|----------|-----------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
|          |                 | B=1          | B=2          | B=3          | B=4          | B=5          | B=1          | B=2          | B=3          | B=4          | B=5          |
| Pubmed   | Degree          | 14.63        | 16.70        | 24.33        | 22.75        | 27.58        | 10.61        | 15.14        | 21.24        | 18.18        | 24.95        |
|          | Page Rank       | 13.51        | 14.48        | 20.44        | 16.86        | 21.10        | 8.47         | 9.56         | 15.71        | 13.73        | 18.31        |
|          | Random          | 2.80         | 5.23         | 7.63         | 11.71        | 13.69        | -2.60        | -8.11        | -9.59        | -12.71       | -14.39       |
|          | Viking          | 2.06         | 5.53         | 8.00         | 11.37        | 15.40        | -2.83        | -5.88        | -10.21       | -13.77       | -13.00       |
|          | Link Prediction | 4.81         | -0.68        | 2.41         | -0.48        | 4.83         | -2.61        | -19.06       | -26.09       | -34.23       | -39.85       |
|          | Nag-R           |              |              | OOT          |              |              | 7.83         | 12.14        | 16.17        | 20.58        | 23.02        |
|          | Nettack         | 1.94         | 3.54         | 6.33         | 9.98         | 14.06        | -6.03        | -12.87       | -21.43       | -28.69       | -30.06       |
|          | <b>Proposed</b> | <b>23.32</b> | <b>38.79</b> | <b>53.18</b> | <b>66.28</b> | <b>76.83</b> | <b>37.78</b> | <b>57.68</b> | <b>69.99</b> | <b>77.53</b> | <b>81.12</b> |
| Amazon   | Degree          | 3.77         | 3.68         | 6.44         | 6.60         | 8.82         | 2.13         | 3.41         | 5.96         | 5.82         | 7.46         |
|          | Page Rank       | 2.99         | 3.04         | 5.71         | 5.21         | 6.23         | 2.23         | 2.81         | 5.04         | 5.52         | 6.53         |
|          | Random          | 0.60         | -0.17        | 0.77         | 1.45         | 0.74         | 0.18         | 0.63         | 1.02         | 0.77         | 0.88         |
|          | Viking          | 0.23         | 0.20         | 0.59         | 0.32         | 0.65         | 0.04         | 0.30         | 0.77         | 0.33         | 0.79         |
|          | Link Prediction | -1.21        | -2.11        | -3.45        | -4.60        | -5.66        | 1.29         | 1.37         | 2.10         | 1.99         | 2.47         |
|          | Nag-R           | 0.22         | 0.54         | 0.98         | 1.55         | 2.11         | 1.59         | 3.00         | 4.15         | 5.51         | 6.56         |
|          | Nettack         | 0.02         | 0.88         | 1.69         | 2.05         | 2.86         | -0.38        | -0.73        | -0.41        | -0.76        | -0.45        |
|          | <b>Proposed</b> | <b>5.73</b>  | <b>9.06</b>  | <b>11.18</b> | <b>13.53</b> | <b>15.34</b> | <b>5.18</b>  | <b>7.46</b>  | <b>9.42</b>  | <b>10.88</b> | <b>12.77</b> |
| CoraFull | Degree          | 4.89         | 4.59         | 7.29         | 8.38         | 10.88        | 2.84         | 4.24         | 8.02         | 10.45        | 13.20        |
|          | Page Rank       | 4.87         | 4.51         | 7.00         | 7.14         | 9.40         | 1.71         | 4.71         | 6.98         | 9.39         | 12.83        |
|          | Random          | 0.80         | 2.45         | 2.21         | 2.43         | 3.06         | 1.36         | 2.64         | 3.11         | 3.02         | 4.26         |
|          | Viking          | 0.98         | 0.20         | 1.61         | 4.45         | 2.86         | 0.69         | 0.26         | 2.79         | 1.72         | 4.91         |
|          | Link Prediction | 1.07         | 1.33         | 2.21         | 2.68         | 3.67         | 0.24         | 1.78         | 3.20         | 3.42         | 5.06         |
|          | Nag-R           |              |              | OOT          |              |              |              |              | OOT          |              |              |
|          | Nettack         | 0.68         | 1.37         | 2.85         | 2.78         | 2.61         | 3.74         | 6.47         | 8.72         | 13.92        | 17.97        |
|          | <b>Proposed</b> | <b>5.88</b>  | <b>9.37</b>  | <b>12.25</b> | <b>14.76</b> | <b>17.02</b> | <b>11.42</b> | <b>17.36</b> | <b>22.29</b> | <b>25.46</b> | <b>28.26</b> |

matrices, full-graph gradient computation despite local perturbations, and sequential per-node attacks. Conversely, our method scales linearly with the target node degree and the number of sampled subgraphs, making it substantially more practical. Together with the complexity analysis in Section 2.4, this addresses **RQ2** by showing superior scalability over existing black-box baselines.

(3) Our method maintains strong performance as the perturbation budget increases, with effectiveness that persists and often strengthens beyond single-edge attacks. This indicates that the scoring mechanism not only identifies impactful edges but also selects deletions that act synergistically. In contrast, baseline methods tend to plateau or show marginal gains at higher budgets due to redundant or suboptimal selections. The widening performance gap as the budget grows highlights our method’s ability to adaptively identify structurally critical edges for demotion, validating **RQ3**.

## 4 Concluding Remarks

We introduced a black-box graph poisoning attack that demotes target node rankings via edge removal without model access. Using sampled subgraphs and local ablation signals to model edge influence, our method consistently outperforms heuristic and gradient-based baselines, scales to large graphs, and improves with higher perturbation budgets. These results demonstrate its effectiveness, scalability, and robustness in practical black-box settings.

## References

- [1] Seong-Jin Ahn and Myoung Ho Kim. 2021. Variational Graph Normalized AutoEncoders. In *CIKM '21: The 30th ACM International Conference on Information and Knowledge Management, Virtual Event, Queensland, Australia, November 1–5, 2021*. ACM, 2827–2831. doi:10.1145/3459637.3482215
- [2] Aleksandar Bojchevski and Stephan Günnemann. 2018. Deep Gaussian Embedding of Graphs: Unsupervised Inductive Learning via Ranking. arXiv:1707.03815 [stat.ML] <https://arxiv.org/abs/1707.03815>
- [3] Yankai Chen, Yixiang Fang, Yifei Zhang, and Irwin King. 2023. Bipartite Graph Convolutional Hashing for Effective and Efficient Top-N Search in Hamming Space. In *Proceedings of the ACM Web Conference 2023, WWW 2023, Austin, TX, USA, 30 April 2023 – 4 May 2023*. ACM, 3164–3172. doi:10.1145/3543507.3583219
- [4] Viresh Gupta and Tanmoy Chakraborty. 2021. VIKING: Adversarial Attack on Network Embeddings via Supervised Network Poisoning. In *Advances in Knowledge Discovery and Data Mining – 25th Pacific-Asia Conference, PAKDD 2021, Virtual Event, May 11–14, 2021, Proceedings, Part III (Lecture Notes in Computer Science, Vol. 12714)*. Springer, 103–115. doi:10.1007/978-3-030-75768-7\_9
- [5] Radin Hamidi Rad, Ebrahim Bagheri, Mehdi Kargar, Divesh Srivastava, and Jarek Szlichta. 2025. Robust Neural Model for Searching over Incomplete Graphs. *ACM Trans. Intell. Syst. Technol.* (May 2025). doi:10.1145/3735650 Just Accepted.
- [6] Amir Khosrojerdi, Amin Bigdeli, Radin Hamidi Rad, Morteza Zihayat, Charles L. A. Clarke, and Ebrahim Bagheri. 2025. Datasets for Supervised Adversarial Attacks on Neural Rankers. In *Proceedings of the 34th ACM International Conference on Information and Knowledge Management, CIKM 2025, Seoul, Republic of Korea, November 10–14, 2025*. ACM, 6432–6437. doi:10.1145/3746252.3761626
- [7] Johannes Klicpera, Aleksandar Bojchevski, and Stephan Günnemann. 2019. Predict then Propagate: Graph Neural Networks meet Personalized PageRank. In *7th International Conference on Learning Representations, ICLR 2019, New Orleans, LA, USA, 2019*. <https://openreview.net/forum?id=H1gL-2A9Ym>
- [8] Yuri Nesterov. 1983. A method for unconstrained convex minimization problem with the rate of convergence  $O(1/k^2)$ . In *Dokl. Akad. Nauk. SSSR*, Vol. 269. 543.
- [9] Radin Hamidi Rad, Silviu Cucerzan, Nirupama Chandrasekaran, and Michael Gamon. 2024. Interactive Topic Tagging in Community Question Answering Platforms. In *Advances in Information Retrieval – 46th European Conference on Information Retrieval, ECIR 2024, Glasgow, UK, March 24–28, 2024, Proceedings, Part III*, Vol. 14610. Springer, 195–209. doi:10.1007/978-3-031-56063-7\_13
- [10] Radin Hamidi Rad, Amir Khosrojerdi, and Ebrahim Bagheri. 2025. ROKSANA: An Open-Source Toolkit for Robust Graph-Based Keyword Search. In *Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval, SIGIR 2025, Padua, Italy, July 13–18, 2025*. ACM, 3994–3998. doi:10.1145/3726302.3730154
- [11] Prithviraj Sen, Galileo Namata, Mustafa Bilgic, Lise Getoor, Brian Gallagher, and Tina Eliassi-Rad. 2008. Collective Classification in Network Data. *AI Mag.* 29, 3 (2008), 93–106. doi:10.1609/AIMAG.V29I3.2157
- [12] Oleksandr Shchur, Maximilian Mümme, Aleksandar Bojchevski, and Stephan Günnemann. 2019. Pitfalls of Graph Neural Network Evaluation. arXiv:1811.05868 [cs.LG] <https://arxiv.org/abs/1811.05868>
- [13] Shu Zhao, Wenyu Wang, Ziwei Du, Jie Chen, and Zhen Duan. 2023. A Black-Box Adversarial Attack Method via Nesterov Accelerated Gradient and Rewiring Towards Attacking Graph Neural Networks. *IEEE Trans. Big Data* 9, 6 (2023), 1586–1597. doi:10.1109/TBDDATA.2023.3296936
- [14] Daniel Zügner, Amir Akbarnejad, and Stephan Günnemann. 2019. Adversarial Attacks on Neural Networks for Graph Data. In *Proceedings of the Twenty-Eighth International Joint Conference on Artificial Intelligence, IJCAI 2019, Macao, China, August 10–16, 2019*. ijcai.org, 6246–6250. doi:10.24963/IJCAI.2019/872